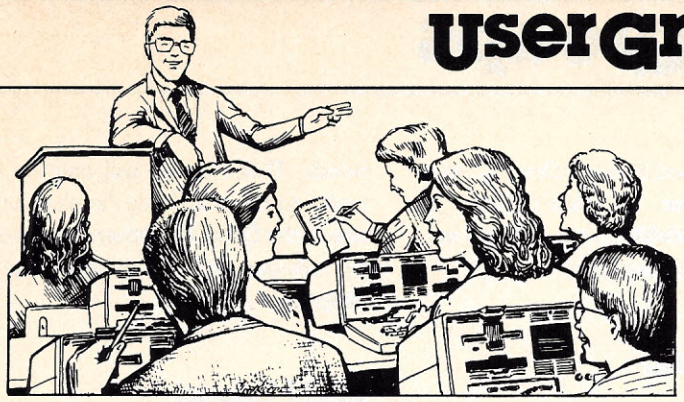
UserGroups



Review of New Drive C: Loader & Archive Programs

Larry Walker

Drive C: came on the market in the summer of 1983 as the answer to the prayer of some of us who were frustrated by the slowness of the Osborne 1 in doing searches in long files, whether in WordStar, dBASE II, SuperCalc or Personal Pearl. Here was a product, we were told, that would look like a disk drive to the Osborne 1 but would act with the speed of random access memory (RAM).

Drive C: was available in two sizes, 192K and 384K. I bought the 384K version because I wanted to load WordStar and Spellguard (with its 50,000 word dictionary) into the Drive C: and still have enough room for writing and editing long documents. (WordStar works best when the maximum document length is no more than one third of the space remaining on the working disk.)

I received my unit in early September 1983 and easily got it up and running. It came with a single floppy disk, which included loading and archiving programs, as well as a 17-page User's Manual.

Much has been written in user group newsletters (some of which has been reprinted in the *Portable Companion* — see April/May 1984 issue, review by Don Kranz, and commenting letters in July 1984 issue) about the inadequacies of the software and the manual. The folks at Drive C: have heard their customers and developed new programs to load the Drive C: and archive and retrieve its contents, and written a substantial user's manual to assist them in making the most of their hardware.

The new software/manual package is being furnished without additional charge to customers who bought their Drive C: units after December 31, 1983. The new package is also available for $39.00 to customers who bought before that date.

This review will be concerned only with the new software and manual and whether they meet their stated goal of making Drive C: easy to use.

For readers who neither own nor have access to a Drive C: unit, a brief description is in order. On the Osborne 1, the unit is mounted in the pocket below the right disk drive. It has two cables emerging from its front panel: the longer one attaches to the IEEE 488 connector on the left side of the Osborne 1, the shorter to the External Video port immediately below the Drive C:. Both of these connectors on the Osborne 1 are duplicated on the panel of the Drive C: so you can connect your Centronics printer cable and shorting plug to it rather than the Osborne 1.

The Drive C: has its own external video jack, so no adapter is needed. In operation, once the Drive C: is loaded (i.e., activated with the loader software), it acts precisely like a third disk drive with the following exceptions: it has no moving parts and thus makes no noise; it works much faster; its memory is volatile, i.e., its contents are lost when power is removed; files to be worked on and programs to be used by Drive C: must be copied into it before it can do any work; you cannot use COPY.COM to copy a disk into Drive C:; when work has been completed, it may be saved to floppy disk for permanent storage.

A number of programs are included in the supplied software. I will quickly summarize these now but explore them further in connection with the review of the manual.

DCL.COM is the Loader program that makes the Drive C: unit accessible. It also allows activation of the print buffer feature. When the buffer is activated, DCL copies DCU.COM, a utility program that controls the use of the buffer, into Drive C:. DCL replaces DRIVEC.COM, the former Loader.

DCA.COM is the Archive-Retrieve program that allows you to save all or any portion of the files contained in the Drive C: unit onto one or more disks and rapidly retrieve them at a later time. This replaces both DCARCH.COM and DCRET.COM.

NOTE:
The original Archiver was called DCARCH.BAS and required the user to load MBASIC before use. This is the version I received; it was *very slow.* It was replaced after a few months with DCARCH.COM, a compiled version of the former program that was a little faster but contained bugs that arose when it was called upon to close files. Thus, some owners stopped using it altogether.

DCQ.COM, DCQINS.COM, DCQ.DAT and DCQAUT.COM constitute the QuickPac package of programs. DCQ.COM is the quick-load program that allows you to store a series of commands in SUBMIT fashion to load Drive C:, name it, copy the contents of a disk (or disks) into it, select a print buffer option, and run a program at the end of the process.

The options chosen for DCQ to run are selected by running DCQINS, a menu-driven Install program, which

saves your choices to disk in the form of an ASCII file called DCQ.DAT. (An initial version of this file is included to get you started. Run the Install program to change any of the options to suit yourself.)

Starting up is made automatic by changing the name of the supplied DCQAUT to AUTOST.COM. A substitute console command processor (CCP), DFD.SPR is designed to make Drive C: work better and is automatically loaded. Finally a print test file, PRN.TST, is included.

The manual runs almost 150 pages, including index, and explains the operation of each of these programs and teaches you how to use them in an integral tutorial section.

Features

The first chapter of the manual previews its contents and sets forth the symbols used throughout. For example, it tries to avoid the confusion that is bound to arise between the name of the unit and the logical name of the drive that has been assigned the name "C" by calling the former "Drive C" and the latter "drive C:". In reading the manual and especially in working through the tutorial, great care should be taken to observe this distinction.

The chapter closes with a brief description of the features of the Drive C: system, which include: higher speed; greater storage capacity; larger maximum file size; ability to designate either of the floppy disk drives or the Drive C: unit as drive A (and to assign "B" or "C" to either of the other two); ability to save the entire contents of the Drive C: unit onto one or more disks and retrieve them quickly from the disk(s) through its Archive-Retrieve program; a print buffer capability that uses either a fixed or variable portion of the Drive C: unit's memory; an automatic rapid-loading system known as "QuickPac"; and the ability to use the Drive C: as a cache buffer/print buffer with a Trantor hard disk, which may be purchased through Drive C: at a discounted price. (I didn't have a Trantor unit available to me so I was unable to review this feature.)

Getting Started

The second chapter presents detailed instructions for mechanical installation of the Drive C: unit in your Osborne 1 and proper hookup of cables as well as trouble-shooting procedures to follow in case the installation does not work as it was supposed to.

It is *very important* that these instructions be followed to the letter (although they are not at all difficult to follow). You are then instructed to prepare a working copy of the Drive C: software disk (the "user disk") and to place the CP/M system tracks on it through the SYSGEN program.

Next comes the printer installation section of the chapter. Some users have had trouble in the past that was traceable to either printer cables that did not meet OCC's specifications or printers such as my Daisywriter 2000 that look "funny" to the Drive C: unit. This section allows you to determine whether your cable is properly wired and explains how to modify it if it is not. The "fix" for using the Daisywriter is either to have the cable disconnected or to have the printer turned ON whenever the Drive C: is in use. (If the printer is connected but OFF, the Loader will not work and the Osborne 1 will crash.)

At the end of the installation procedures, PRN.TST is sent to the printer; it will print a proper test pattern when printer installation is proper. I believe that many cries for help will be averted if the user carefully follows this chapter, step by step.

Methods of Use

The third chapter briefly tells how to use the loader software in QuickPac and/or manual modes to best suit your needs. This chapter will only be meaningful to those already familiar with the use of the Drive C:. Others would be better advised to read it after the sixth chapter.

CP/M Tutorial

The fourth chapter is devoted to a short tutorial on some of the simple CP/M commands and error messages. This is directed at users who have no working knowledge of CP/M's PIP, XDIR, REN, and ERA commands, because these commands are used heavily in the following chapters.

Many users will be able to skip this, but if you are a novice you would be well advised to take the few minutes required to work your way through this chapter.

Drive C: Tutorial

We finally get to the meat of the manual in the fifth and sixth chapters, which take the user, step by step, through all of Drive C:'s features. In the course of this tutorial you prepare a blank floppy "demo disk" and, in conjunction with the "user disk" you prepared in chapter two, learn to use the various CP/M commands and the Drive C: software to: install Drive C: and give it logical name A, B or C; move files into and out of the Drive C: unit (WordStar is used for demonstration purposes); restart use of the Drive C: unit after pressing the reset button (which does *not* destroy the contents of the unit; only turning off the computer does that); erase all files contained in the unit (by using the DCU E command); and use the print buffer and observe it in action (the buffer allows you to use the computer while the buffer drives the printer).

Chapter 5 is a well-presented walkthrough of all of these features. It is a far cry from the three pages of the original manual devoted to the subject matter (although the original software did not provide the user with print buffer capa-

bility, which constitutes half of the content of this chapter).

The new software is much easier to use than the prior version. Where, for example, you had to type `DRIVEC A: C:↵` to activate Drive C: and name it logical A and the left floppy logical C, you now have only to type `DCL A↵`.

Also, you can now rename the Drive C: unit at any time by typing the command `DCN x↵` at the CP/M prompt, where `x` is the new logical name of the Drive C: unit. (The DCN.COM file is automatically generated and loaded into Drive C: when the loader program is run.)

The print buffer software is also loaded at the time the loader program is run. Using the above example, if `DCL A` is followed by any of the numbers, `16, 32, 48, 64, 80, 96, 112,` or `128` in the form `DCL A 16↵`, 16K (or whatever K) of the Drive C:'s memory is set aside as a print buffer and is unavailable as a program processer. Choosing the letter `P` instead of one of those numbers makes available for print buffer service all otherwise unused memory (the dynamic buffer option), in which case, dumping the contents of the buffer into the printer automatically releases this memory space for program processing. Any other number or letter will be ignored and no buffer will be installed.

Finally, the buffer utility program (DCU.COM) is loaded into the Drive C: unit whenever the print buffer is installed. When `DCU↵` is typed, this program presents a menu offering a number of options for controlling the buffer: you can stop and restart the buffer-to-printer dump, clear the buffer, determine the space available for the dynamic buffer, or even cause the files in the Drive C: to be compressed to make greater space available for the dynamic buffer.

It should be noted that none of these controls are likely to be available if you already have a buffer (outboard or built into the printer) between your Osborne 1 and your printer, unless your file is larger than the buffer, because the Drive C: will probably dump its contents into the buffer faster than you can move to give it a command. You might just want to sell your buffer; it would reduce the cost of your Drive C:.

QuickPac Tutorial

This constitutes the first part of the sixth chapter. As mentioned earlier, the QuickPac package of programs is designed to allow the user to start the Loader automatically and copy the de-

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sired files into the Drive C: unit as quickly as possible. There are several ways of doing this, two of them are faster and more automatic than the others.

When running the Install program you are first asked if you want to copy any disks to Drive C:. If you answer `Y`, you are given the option of copying all the files from the left, right or both floppy drives to Drive C:. This copying procedure uses the PIP command, which is a relatively slow file-copying program.

If you answer `N` to the copy question, you can instead automatically retrieve previously archived sets of files from one or two disks. This is done by utilizing the feature that allows you, at the end of the Install procedure, to designate a program to be run. If you designate `DCA R B` ↵ as the desired program, QuickPac will run the Archive-Retrieve program in the Retrieve mode and look to drive B: (the right floppy drive) for the archived disk(s). (The above assumes you have chosen logical drive A: as the name of the Drive C: unit. The DCA command would differ if another name were chosen.)

The retrieve program is *very fast*, copying WSMSGS.OVR, a 34K file, to Drive C: in just 10 seconds. Compare that to PIP, which takes 17 seconds. My set of WordStar/Spellguard related files totaling 272K loads in 115 seconds. Single density owners will find that the process would take a little longer due to the need to change disks twice.

As stated earlier, the Install program asks you a number of questions besides those relating to copying files and running programs. For example, it asks you what logical name you want to give to Drive C:; it asks whether you want to install the print buffer and if so what size; and it asks on which drive you want to store the DCQ.DAT file.

The only question asked by Install that might give trouble is the name of the drive on which you want to store the file DCQ.DAT. Remember, this is the file which stores your answers to all the questions asked by the Install program. The drive you choose should be the one with the Drive C: user's disk, as it is *then* named (at the time you are running the Install program). If you are running the Install program without having first loaded Drive C:, the left floppy is still drive A: and your answer should be `A`.

On the other hand, if you have already loaded Drive C: and are running the Install program to modify your load routine, your user disk is probably not on drive A: anymore. (This assumes that you have named Drive C: something other than `C`.)

If you give the wrong answer, QuickPac will look to the wrong disk for the DCQ.DAT file. If it finds no such file on that disk, it will give you the cryptic error message `DCQ.DAT?`. If it finds a file of that name on the wrong disk, it will doubtless be the wrong version of this file and QuickPac will not do the things you want it to. This error can be a pain to track down, but the manual provides trouble-shooting procedures.

Sometimes when you are working with Drive C: you will need to use your Osborne 1 for another task. As mentioned earlier, you can push the reset button without losing Drive C:'s contents. (For safety's sake, remember to save and back up your work before pushing the button.) When you want to return to your original project, it is a simple matter to insert your user disk and start QuickPac again. If you have been using the copy option, the Loader will not try to copy files that are already contained in Drive C:.

On the other hand, if you have chosen to run DCA to retrieve archived files as part of your QuickPac routine, the DCA program *will* insist on retrieving your files again. A good way to avoid this is to have another "continue" user's disk, which, instead of running `DCA R B` ↵ at the end of the quick-load procedure, runs, for example, a program like `XDIR`.

Archive Tutorial

The rest of the sixth chapter is devoted almost entirely to the archiving to and the retrieving from disk of files stored in Drive C:. The tutorial demonstrates three procedures for saving files to floppy disk: using the "cOpy" command from the WordStar no file menu (similar saving procedures are available in the other applications programs, such as Supercalc and dBase II); using PIP, complete with CP/M "wildcard" designators; and using DCA.COM.

When I am working in WordStar and periodically save my work with a Control-`KD` command, I also back it up to floppy disk using the "cOpy" command. At other times I find the convenience of PIP and its wildcard capabilities to be preferable.

The Archive program is best when the same group of files will be used again and again, because I don't have to copy anything to disk except files that have been changed permanently. (Incidentally, much of the pain of disk-changing with Personal Pearl can be avoided by archiving a set of disks containing all the programs required to add or edit data, produce reports, and sort the data, totaling 256K. Using Drive C: with Pearl speeds up the sorting and report production process substantially.)

Convenience of storage using the Archive program is enhanced by the menu, which allows you quickly to select (and unselect) files to store, display a directory of all files in Drive C:, and list the files already selected for storage.

The program also has built-in safety features to prevent saving files onto disks that are not blank by warning you of this fact and displaying a directory of the files on the disk in question. It then asks you twice whether you really want to erase all those files before starting to store files to that disk.

When archiving, all the files are stored contiguously as one file on the disk. These files are only accessible through the Retrieve program. If you look at a directory of the disk it will show `filename.AR1` only. (You assign the filename by answering the appropriate question on the Archive menu.) If you are loading files that take up more than

one disk of space, the program instructs you when to change disks, and it assigns the second disk the name `filename.AR2`.

I am told by Dave Price, the CEO of Drive C:, that the speed of the retrieval process already commented on is directly attributable to the method of data storage, but he didn't get more specific than that; and it's just as well considering my lack of technical knowledge of computers.

From the above, it may be seen that the Archive-Retrieve program is easy to use and an efficient method of storing files to disk. In my view it is one of the best features of the new software. It stands head and shoulders above the former software.

BackPac

The folks at Drive C: have developed a battery pack that installs within the keyboard enclosure and keeps the memory of the Drive C: unit active for 8-10 hours after the power goes off, deliberately or through power failure. I am told that the manual for this unit is being written and the product will not be released until it is completed. Target price is around $150. I hope to review this product when it is available.

Conclusion

The new software greatly enhances the use of an already fine piece of hardware. It should enable even the tyro to make very good use of the Drive C: RAM-disk. Its developers are to be congratulated. Also, it is important to note that the people at Drive C: are truly dedicated to making the Osborne 1 and Osborne Executive computers more valuable to their owners. The support they give to their customers is, in my experience, unprecedented. In fact, but for this high level of support, the new software/manual might have been available weeks if not months earlier. Drive C: is available from Drive C:, 1690 65th Street, Emeryville, CA 94608.

Larry Walker is a member of East Bay FOG. This article was reprinted from FOGHORN *(First Osborne Group, Daly City, CA), 9/84.*

Commx-Pac: A Smart Terminal Program for Osborne and CP/M Computers

Alan R. Bechtold

Continued on page 62

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